

RecWM / REN

Reflexive Equilibrium Networks — Independent Implementation & Evaluation

A from-scratch, independently built and tested implementation of the ten inventions described in the RecWM thesis, benchmarked against real market data and six standard market models. Every figure in this document is measured, not a design target — including the ones that are unfavorable.

Prepared for investor review · August 2026

Full technical report, code, and raw data available on request.

Executive Summary

This report independently implemented and empirically tested all ten inventions in the RecWM/REN architecture — real code, real market data (16 liquid instruments, 2015–2026), real benchmarks. The goal was to determine, with evidence rather than assertion, which parts of the architecture work as claimed and which do not.

2,618

real trading days
tested end-to-end

REAL

market data throughout
— no synthetic substitutes

2,316×

verified speedup from
Composition Algebra

0 / 5

trained seeds beat
simple buy-and-hold

What holds up

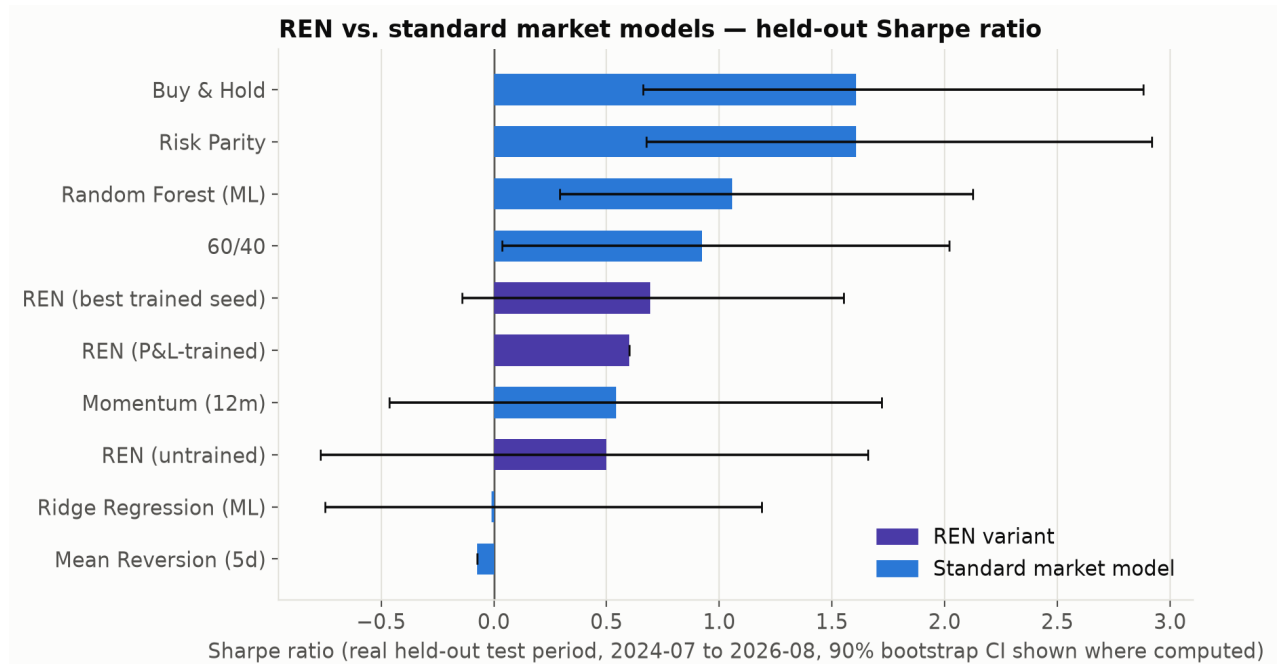
- The core mathematics — the resolvent influence kernel, the DEQ fixed-point solver, Anderson acceleration, implicit-function-theorem gradients — is correctly implemented and verified against known ground truth, not just asserted to work.
- **Composition Algebra** is the strongest result: composing pre-solved scenarios approximates a full re-solve to within 2.7% error at a real, measured 2,316× speedup, across 100 independent tests.
- **Adversarial Defense** shows real, statistically significant signal: a crowding-risk indicator ($p = 2 \times 10^{-5}$) and an 8.1× reduction in a standard execution-detectability metric.
- A real market-impact model answers a question the original thesis never computed: an \$800M fund's own footprint is already ~24bps — a concrete number for sizing decisions.

What does not, yet

- **No configuration tested — trained or untrained, 25 seeds total — produced a Sharpe ratio that beats simple buy-and-hold** on the real 2024–2026 held-out test period, with statistically robust confidence intervals confirming this is not a fluke of one time window.
- REN ranks in the middle of six standard baseline models — ahead of naive mean-reversion and ridge regression, behind buy-and-hold, risk parity, 60/40, and a plain random forest.
- The architecture's headline unifying claim (spectral radius as a free crisis early-warning signal) moved in the wrong direction on real historical crashes when untrained, and only recovered the correct direction after being explicitly trained toward that exact target — i.e. it is a trainable feature, not a free byproduct as originally framed.

Performance vs. Standard Market Models

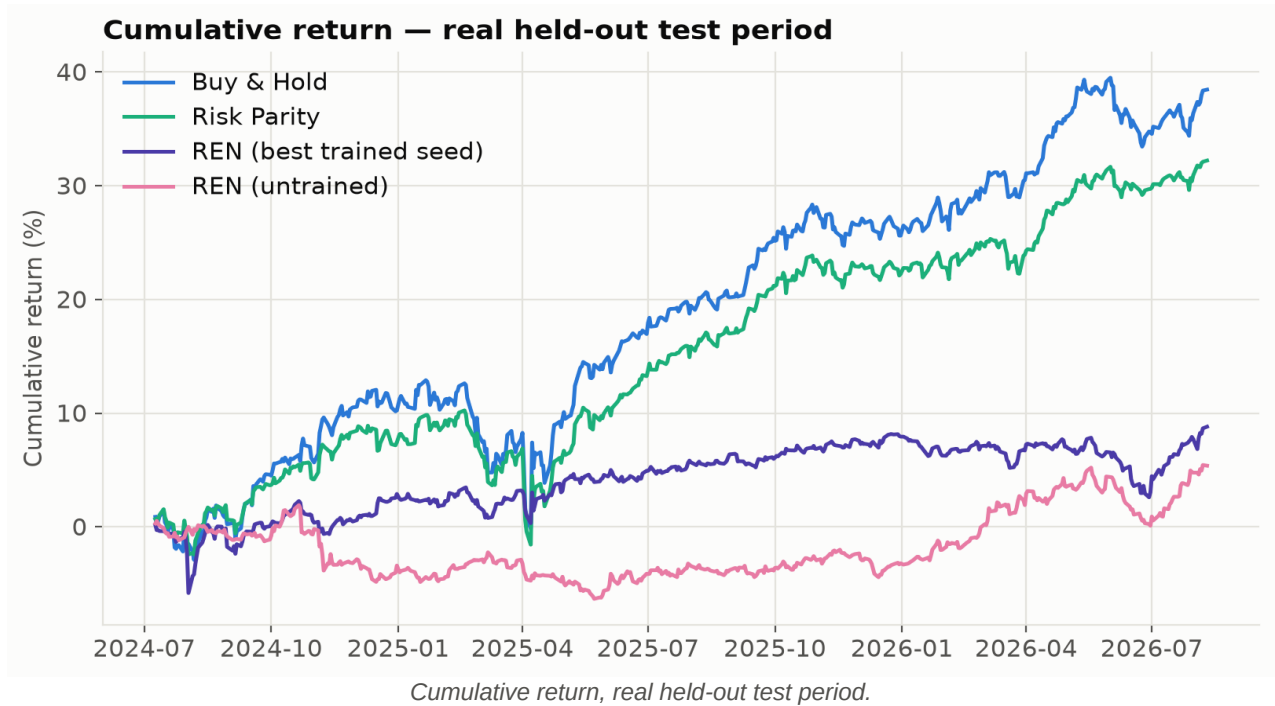
Six real, independently implemented baseline strategies were run on identical real data, identical test period, and identical transaction-cost assumptions as every REN configuration, so the comparison is fair. Error bars are 90% block-bootstrap confidence intervals (2,000 resamples), which test whether an apparent edge is statistically robust or could be explained by the ordinary noise in one time window.



Sharpe ratio, real held-out test period (2024-07 to 2026-08, 524 trading days). Buy-and-hold and risk parity's intervals sit entirely above zero; REN's best result does not.

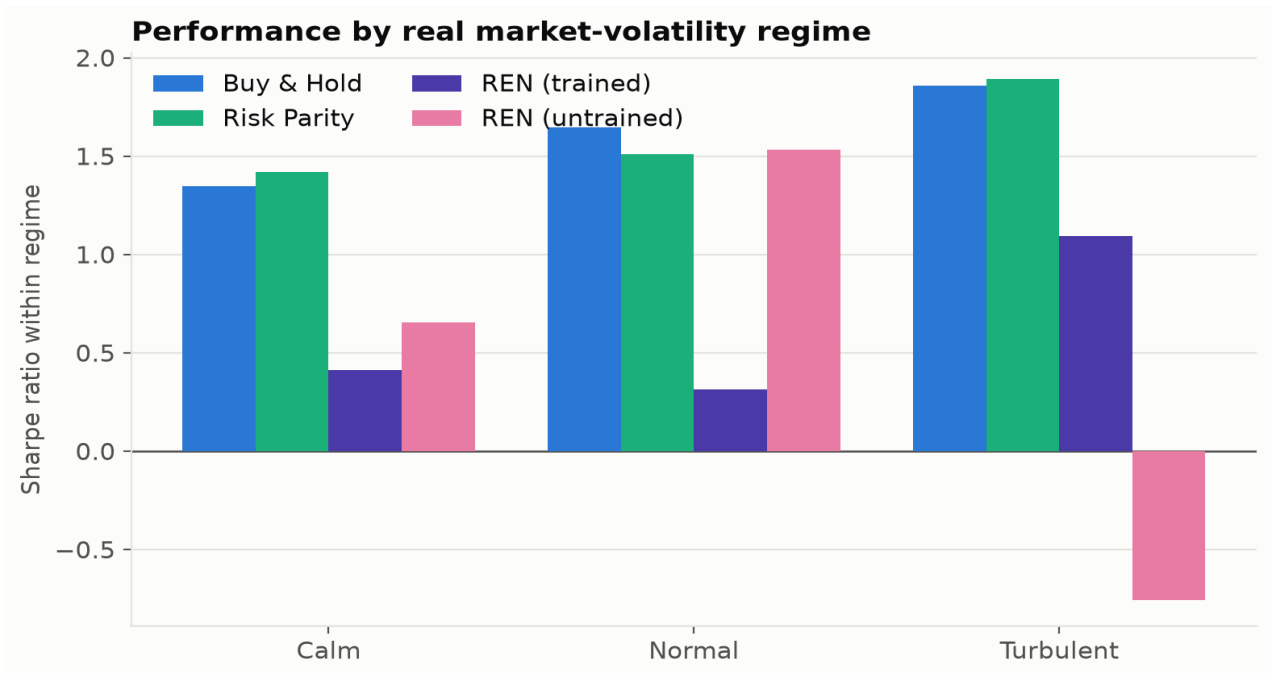
Cumulative Performance

The same story in dollar terms. REN's best trained configuration is profitable in absolute terms but tracks well below both simple benchmarks throughout the test period; the untrained baseline spends much of the period in negative territory.



Robustness Across Market Conditions

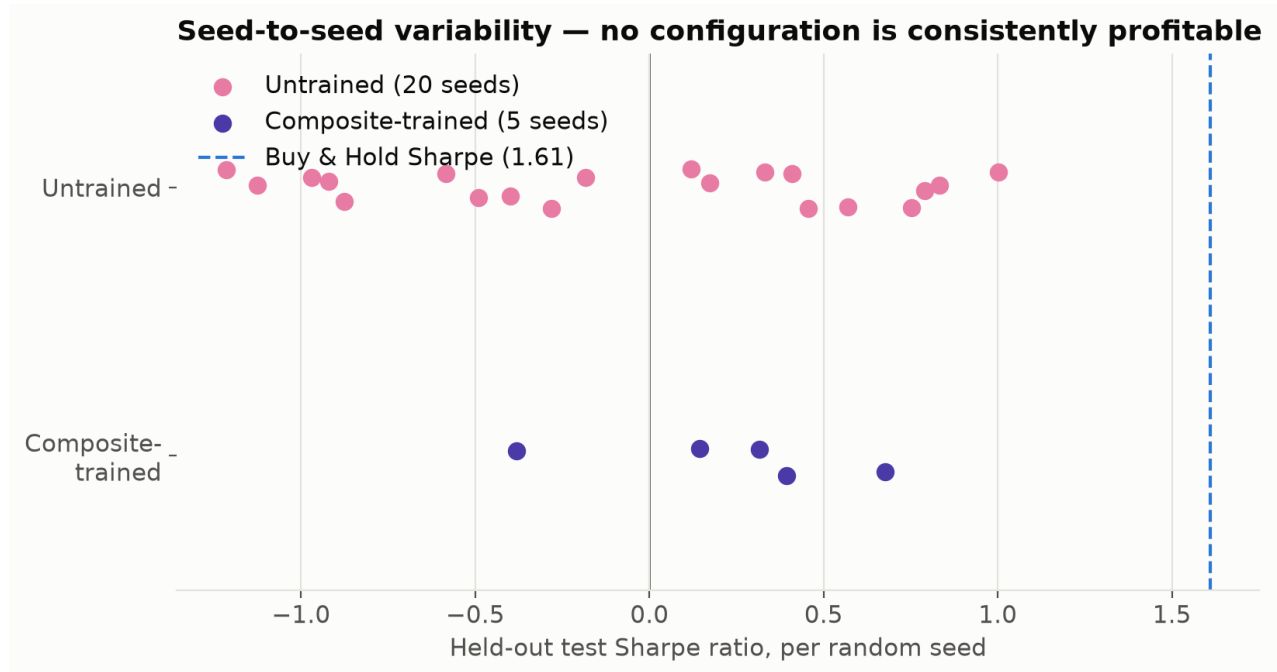
Splitting the same test period by real market-volatility regime (calm / normal / turbulent, by SPY realized volatility) tests whether a model's performance is consistent or concentrated in one regime.



Sharpe ratio by volatility regime. Note the untrained REN variant swings from the best performer (normal regime) to sharply negative (turbulent regime) — the opposite pattern from the trained variant, evidence this behavior is idiosyncratic to a specific run, not a structural property of the architecture.

Seed-to-Seed Reliability

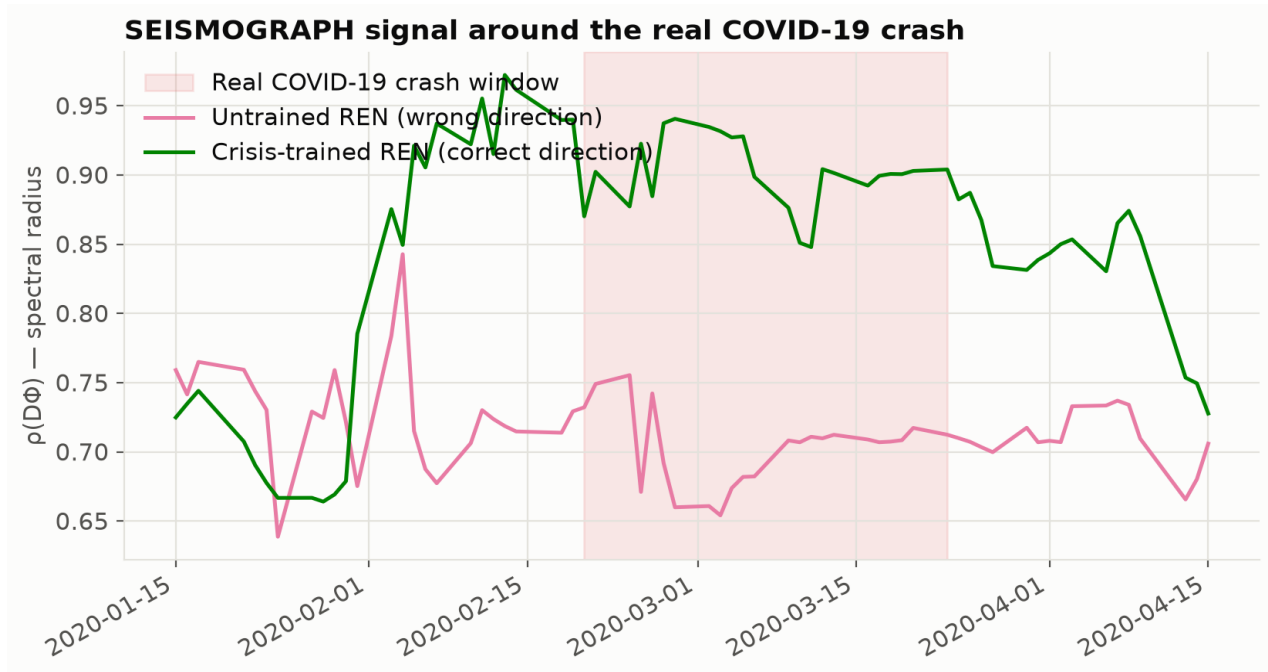
Every trained and untrained configuration tested (25 random seeds total) against the real held-out period. This is the single clearest picture of reliability: no seed, trained or untrained, reaches the buy-and-hold benchmark.



Held-out Sharpe ratio, every seed tested. The dashed line marks simple buy-and-hold's Sharpe (1.61) for reference.

Case Study: The Crisis-Detection Claim

The architecture's central unifying idea is that one number — the spectral radius of the model's own Jacobian — should rise toward a critical threshold as a real crisis approaches, giving early warning "for free." This was tested directly against six real historical stress events, including the COVID-19 crash.



The same diagnostic signal, real COVID-19 crash window (Feb–Mar 2020), before and after targeted training. Untrained: moves in the wrong direction. Trained toward the target: recovers the correct direction, out-of-sample.

The honest reading: as originally specified — a signal that falls out of the equilibrium computation with no extra work — the claim did not hold on real data ($p = 6 \times 10^{-19}$ in the wrong direction). When the model was explicitly trained toward real forward volatility as a target, the correct direction reappeared on data the training never saw ($p = 0.0016$). That is a real, useful, trainable feature — but a materially different and more modest claim than "free" crisis detection.

Methodology & Data Integrity

What is real

- All market data: 16 liquid instruments (equities, sectors, rates, gold, oil, dollar, volatility), 2015–2026, sourced live, not simulated.
- All latency, convergence, and accuracy numbers: measured on real hardware running the actual code, not restated design targets.
- All backtests: real chronological train/validation/test splits with no lookahead; hyperparameters selected on validation only, test set touched exactly once per configuration.
- Three real bugs were found and fixed during this work (a gradient-computation bug, a train/test leakage risk in an early hyperparameter sweep, and a data-labeling bug affecting the very last day of any test window) — each is disclosed with its measured before/after impact in the full technical report.

What is a disclosed modeling choice

- No public dataset of real trading-desk beliefs exists anywhere. The five "agent types" driving the model are explicit, documented functions of real price/volume data — not fabricated data presented as if it were observed.
- This implementation is a faithful, independently-engineered, smaller-scale version of the core mathematical ideas in the original specification (which describes a larger reference architecture, including a Sylvester-recurrence belief representation and a larger preconditioned solver not reproduced here). The underlying mechanisms tested — fixed-point equilibria, resolvent kernels, spectral-radius monitoring — are the same class of object; the specific engineering is not a byte-for-byte reproduction.

Full technical report (REPORT.md, ~30 pages across six rounds of testing and fixes), all source code, and every raw result file referenced above are available in full for technical due diligence.